

Kim Sanghyeon 김상현

Blockchain Node Operations · Infrastructure

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SUMMARY

Blockchain infrastructure engineer operating Kaia, Avalanche, and Celestia validators. Built an AWS jump server for controlled access and Prometheus/Grafana monitoring with multi-channel alerts. Also researched smart-contract security and reported a prediction-market vulnerability (**\$1,000 bounty**).

EDUCATION

Korea University – School of Cybersecurity – korea.ac.kr

2021.03 – 2027.02 (expected)

Enrolled

EXPERTISE

Focus	Blockchain node operations · Infrastructure monitoring · Smart-contract security
Infra	Linux/RHEL · AWS (EC2·VPC·EIP) · OVH Bare Metal · Prometheus · Grafana · RAID
Security	Solidity · Foundry · pwntools · gdb · IDA/Ghidra · Threat Modeling
Lang	Python · C · Solidity · Git
Cert	Network Administrator Lv.2

EXPERIENCE

Blockchain Validator · Infrastructure Engineer – [Anam145](#)

2026.05 – Present

Full-time · Infra team · Validator operations

- Operate multi-chain validators (**Kaia** · Avalanche · Celestia) and in-house infrastructure
 - Built an **AWS jump server** to remove direct exposure and enforce a single access point
 - Set up **Prometheus·Grafana** monitoring with multi-channel alerts
- Developing an in-house **Web3 AI security auditor**

Information Security Specialist · SOC – *Defense Integrated Data Center*

2022.07 – 2024.01

ROK Army · Monitoring

- Packet analysis with IPS·WAF; blocked malicious mail and IPs.
- Real-time threat detection and response using IDS and security-log tools.

PUBLICATION & TALK

Structural Limits of PQC Migration in Ethereum Smart Contracts – [CISC-S'26](#)

2026.01 – 2026.05

Survey paper · Co-author

- Co-authored a survey of challenges in migrating Ethereum smart contracts to post-quantum cryptography.

.HACK 2026 Conference Talk – [dothack.io](#)

2026.04.08

- A four-topic talk on prediction-market security – off/on-chain latency abuse, single-oracle dependence, 51% oracle-governance attack, and Opinion overflow trade-blocking.

PROJECTS

Web3 AI Security Auditor – *Anam145*

2026.08 – Present

- LLM-automated smart-contract vulnerability auditing, applying patterns from the prediction-market security research.

Prediction-Market Security Guide · Lead – *Upside Academy*

2025.08 – 2026.04

funarchy.gitbook.io/

- **Led the team.** Threat-modeled prediction-market protocols – **21 vulnerability patterns across 5 security domains**, assessing **9 live protocols** (Polymarket, Opinion, ...).
- Found an **integer-overflow vulnerability in Opinion's settlement** that could block a specific user's trades → PoC · report → **\$1,000 bug bounty**.

ACTIVITIES

Blockchain Valley · Research Team – *medium*

2026.03 – Present

- Published research on the trust problem oracles pose when representing **RWAs (real-world assets)** on-chain (*RWA and the Fundamental Contradiction of Oracles*↗).
- Currently on the development team.

Upside Academy – *Web3 Security Bootcamp*

2025.08 – 2025.12

- Dunamu × Theori Web3 security program; built the prediction-market security guide as a team project.

Cykor Hacking Club · Member – *Korea University*

2021.03 – 2022.07

- Studied systems/web/crypto and pwnable techniques (**RTL · ROP**); found vulnerabilities via static/dynamic analysis and IDA/Ghidra reversing in CTFs.
- **Taught RTL · ROP** to junior cohorts.

42 Seoul – *Developer Program*

2021.07 – 2022.06

- Participated in the peer-learning curriculum and studied introductory C programming.